# Derivatives

Last modified: 2026-11-11 09:22:04 (PST)

## Limits and derivatives

**Definition 0.1** (Limit of a function at a point)**.** Let $f$ be a function defined at every point of an open interval around $c$, except possibly at $c$ itself, and let $L$ be a real number. The **limit** of $f(x)$ as $x$ approaches $c$ is $L$ [1], written $\lim_{x\to c} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for every } x \text{ with } 0 < |x - c| < \delta.$$

Here $|\cdot|$ is the absolute value. When such a real number $L$ exists, the limit **exists**; otherwise, the limit does not exist. The value $f(c)$, if it is defined, plays no role.

**Example 0.1** (The limit of $3x+1$ at 2)**.** $\lim_{x\to 2}(3x+1) = 7$. Given $\varepsilon > 0$, take $\delta = \varepsilon/3$. For every $x$ with $0 < |x-2| < \delta$, using the distributive law in the second line,

$$\begin{aligned}
|(3x+1) - 7| &= |3x - 6| && \text{(subtract)} \\
&= |3(x-2)| && \text{(distributive law)} \\
&= 3|x-2| && (|3y| = 3|y|, \text{ since } 3 > 0) \\
&< 3 \cdot \frac{\varepsilon}{3} && (|x-2| < \delta = \varepsilon/3) \\
&= \varepsilon && \text{(multiply)}
\end{aligned}$$

For example, with $\varepsilon = 0.3$ and $\delta = 0.1$, the point $x = 2.05$ satisfies $0 < |2.05 - 2| < 0.1$, and

$$\begin{aligned}
|(3 \cdot 2.05 + 1) - 7| &= |7.15 - 7| \\
&= 0.15 \\
&< 0.3.
\end{aligned}$$

**Definition 0.2** (Right-hand limit)**.** Let $f$ be a function and $L$ a real number. The **right-hand limit** of $f$ at $c$ is $L$ [2], written $\lim_{x \to c^+} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every $x$ with $c < x < c + \delta$.

**Definition 0.3** (Left-hand limit)**.** Let $f$ be a function and $L$ a real number. The **left-hand limit** of $f$ at $c$ is $L$ [2], written $\lim_{x \to c^-} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every $x$ with $c - \delta < x < c$.

**Definition 0.4** (One-sided limits)**.** Let $f$ be a function. The right-hand limit (Definition 0.2) and the left-hand limit (Definition 0.3) of $f$ at $c$ are the **one-sided limits** of $f$ at $c$.

**Example 0.2** (One-sided limits of a step function)**.** Let $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$.

- $\lim_{x \to 0^+} H(x) = 1$, because $H(x) = 1$ for every $x > 0$.
- $\lim_{x \to 0^-} H(x) = 0$, because $H(x) = 0$ for every $x < 0$.

The one-sided limits differ, since $1 \neq 0$, so $\lim_{x \to 0} H(x)$ does not exist.

Remark (A limit exists exactly when both one-sided limits exist and are equal)

*Remark* 0.1 (A limit exists exactly when both one-sided limits exist and are equal). The limit $\lim_{x \to c} f(x)$ (Definition 0.1) exists exactly when both one-sided limits (Definition 0.4) exist and are equal, and then all three are equal.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist introduces limits and derivatives:

- Calculating Limits
- Exercises on Limits
- Intro to Differential Calculus
- How Derivatives Arise from Limits
- Derivative Notation

**Definition 0.5** (Difference quotient)**.** Let $f$ be a real-valued function of one variable, defined at $c$ and at $c+h$, with $h \neq 0$. The **difference quotient** of $f$ at $c$ with increment

$h$ is [3]

$$\frac{f(c+h)-f(c)}{h},$$

the slope of the line through the points $(c, f(c))$ and $(c+h, f(c+h))$.

**Example 0.3** (The difference quotient of $x^2$ at 1)**.** For $f(x) = x^2$, $c = 1$, and $h \neq 0$,

$$\begin{aligned}
\frac{f(1+h)-f(1)}{h} &= \frac{(1+h)^2-1}{h} && \text{(substitute into } f\text{)} \\
&= \frac{1+2h+h^2-1}{h} && \text{(expand } (1+h)^2\text{)} \\
&= \frac{2h+h^2}{h} && (1-1=0) \\
&= 2+h, && \text{(divide by } h \neq 0\text{)}
\end{aligned}$$

which tends to 2 as $h \to 0$ (Definition 0.1). With $h = 0.1$ the difference quotient is $2 + 0.1 = 2.1$: the line through $(1, 1)$ and $(1.1, 1.21)$ has slope

$$\begin{aligned}
\tfrac{1.21-1}{0.1} &= \tfrac{0.21}{0.1} \\
&= 2.1.
\end{aligned}$$

**Definition 0.6** (Differentiable function)**.** A function $f$ is **differentiable at** $x = c$ [4] if the limit (Definition 0.1) of the difference quotient (Definition 0.5) as $h \to 0$,

$$f'(c) \stackrel{\text{def}}{=} \lim_{h \to 0} \frac{f(c+h)-f(c)}{h},$$

exists and is finite. The value of this limit is denoted $f'(c)$.
[5, Sec. 2.1], p. 100

**Example 0.4** (A differentiable function, and a limit that is not finite)**.**

- $f(x) = x^2$ is differentiable at $c = 3$:

$$\begin{aligned}
\frac{f(3+h)-f(3)}{h} &= \frac{(3+h)^2-3^2}{h} && \text{(substitute } f(x)=x^2\text{)}\\
&= \frac{9+6h+h^2-9}{h} && \text{(expand } (3+h)^2\text{)}\\
&= \frac{6h+h^2}{h} && \text{(cancel } 9-9\text{)}\\
&= 6+h && \text{(divide by } h\neq 0\text{)}
\end{aligned}$$

which tends to the finite limit $f'(3) = 6$ as $h \to 0$.

- $g(x) = \sqrt[3]{x}$ is not differentiable at $c = 0$: the difference quotient

$$\begin{aligned}
\tfrac{g(h)-g(0)}{h} &= \tfrac{\sqrt[3]{h}}{h}\\
&= \tfrac{1}{(\sqrt[3]{h})^2}
\end{aligned}$$

grows without bound as $h \to 0$, so the limit is not finite.

**Definition 0.7** (Derivative)**.** If $f$ is differentiable at $c$ (Definition 0.6), the number $f'(c)$ is the **derivative** of $f$ at $c$ [6]. The **derivative** of $f$ is the function $f'$ whose value at each $x$ where $f$ is differentiable is $f'(x)$.

**Definition 0.8** (Lagrange's notation)**.** Writing the derivative (Definition 0.7) with a prime, as $f'$ or $f'(x)$, is **Lagrange's notation** [7].

**Definition 0.9** (Leibniz notation)**.** The derivative of $f$ (Definition 0.7) is also written $\frac{df}{dx}$ or $\frac{d}{dx}f(x)$, which is **Leibniz notation** [7].

**Example 0.5** (The derivative of $x^2$)**.** For $f(x) = x^2$, the computation in Example 0.4, with 3 replaced by any real number $c$, gives $\frac{f(c+h)-f(c)}{h} = 2c + h$, which tends to $2c$ as $h \to 0$. So the derivative of $f$ is $f'(x) = 2x$. For example, $f'(3) = 6$ and $f'(-1) = -2$.

**Definition 0.10** (Right-hand derivative)**.** Let $f$ be a function defined at $c$. The **right-hand derivative** of $f$ at $c$ is the right-hand limit (Definition 0.2) $\lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h}$ of the difference quotient (Definition 0.5), when that limit exists and is finite.

**Definition 0.11** (Left-hand derivative)**.** Let $f$ be a function defined at $c$. The **left-hand derivative** of $f$ at $c$ is the left-hand limit (Definition 0.3) $\lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h}$, when that

limit exists and is finite.

**Definition 0.12** (One-sided derivatives)**.** Let $f$ be a function defined at $c$. The right-hand derivative (Definition 0.10) and the left-hand derivative (Definition 0.11) of $f$ at $c$ are the **one-sided derivatives** of $f$ at $c$.

**Example 0.6** (One-sided derivatives of $|x|$ at 0)**.** Let $f(x) = |x|$, the absolute value, and $c = 0$.

- For $h > 0$, $|h| = h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{h - 0}{h} \\ &= 1.\end{aligned}$$

  The right-hand derivative of $f$ at 0 is 1.

- For $h < 0$, $|h| = -h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{-h - 0}{h} \\ &= -1.\end{aligned}$$

  The left-hand derivative of $f$ at 0 is $-1$.

For example, $h = 0.5$ gives $\frac{0.5}{0.5} = 1$, and $h = -0.5$ gives $\frac{0.5}{-0.5} = -1$. The one-sided derivatives differ, since $1 \neq -1$.

**Definition 0.13** (Second derivative)**.** Let $f$ be a real-valued function of one variable whose derivative $f'$ (Definition 0.7) exists at every point of an open interval containing $c$. If $f'$ is differentiable (Definition 0.6) at $c$, its derivative there, written $f''(c)$ or $\frac{d^2 f}{dx^2}$, is the **second derivative** of $f$ at $c$ [8].

**Example 0.7** (The second derivative of $x^3$)**.** For $f(x) = x^3$, the power rule (Theorem 0.3) gives $f'(x) = 3x^2$ at every $x$, and differentiating again, with the constant multiple rule (Theorem 0.2), gives

$$\begin{aligned}f''(x) &= 3 \cdot 2x \\ &= 6x.\end{aligned}$$

So $f''(2) = 12$, $f''(0) = 0$, and $f''(-1) = -6$.

## Derivative rules

**Theorem 0.1** (Constant rule)**.** *If $c$ is constant with respect to $x$, then*

$$\frac{\partial}{\partial x} c = 0$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Derivative of a Constant

**Theorem 0.2** (Constant multiple rule)**.** *If $a$ is constant with respect to $x$ and $y$ is a differentiable function of $x$, then:*

$$\frac{\partial}{\partial x} ay = a \frac{\partial y}{\partial x}$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Constant Multiple Rule for Derivatives

**Theorem 0.3** (Power rule)**.** *For every real number $q$ and every $x > 0$ [9], the derivative of the power $x^q$ is:*

$$\frac{\partial}{\partial x} x^q = q x^{q-1}$$

*When $q$ is a positive integer, the same formula holds for every real $x$.*

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule, on the sum rule (the derivative of a sum is the sum of the derivatives), and on exercises that combine the rules so far:

- The Power Rule for Derivatives
- The Sum Rule for Derivatives
- Exercises on Derivative Rules

**Exercise 0.1** (Derivative of polynomial and power functions)**.** Adapted from Miller [10], Question 1.1.1.
Find the derivative of

$$f(x) = 4x^5 + 3x^2 + x^{1/3}$$

Solution

*Solution* 0.1. Differentiate term by term using the constant multiple rule (Theorem 0.2) and the power rule (Theorem 0.3):

$$\begin{aligned}
\frac{d}{dx}(4x^5) &= 4 \cdot (5x^4) \\
&= 20x^4 \\
\frac{d}{dx}(3x^2) &= 3 \cdot (2x) \\
&= 6x \\
\frac{d}{dx}(x^{1/3}) &= \frac{1}{3}x^{1/3-1} \\
&= \frac{1}{3}x^{-2/3}
\end{aligned}$$

Combining the three terms yields:

$$f'(x) = 20x^4 + 6x + \frac{1}{3}x^{-2/3}$$

**Exercise 0.2** (Finding points with a prescribed slope)**.** Adapted from Miller [10], Question 1.1.14.
Let $f(x) = \frac{1}{3}x^3 + x^2 - x - 1$. Find all points on the graph of $f$ where the slope is:

1. $-1$
2. $2$
3. $0$

Solution

*Solution* 0.2. The slope is given by the derivative:

$$f'(x) = x^2 + 2x - 1$$

**1. Slope $-1$:** Solve

$$\begin{aligned} x^2 + 2x - 1 = -1 &\iff x^2 + 2x = 0 \\ &\iff x(x+2) = 0. \end{aligned}$$

The roots are $x = 0$ and $x = -2$. Evaluating $f$: $f(0) = -1$ and

$$\begin{aligned} f(-2) &= -\frac{8}{3} + 4 + 2 - 1 \\ &= \frac{7}{3}. \end{aligned}$$

The points are $(0, -1)$ and $(-2, 7/3)$.

**2. Slope 2:** Solve

$$\begin{aligned} x^2 + 2x - 1 = 2 &\iff x^2 + 2x - 3 = 0 \\ &\iff (x+3)(x-1) = 0. \end{aligned}$$

The roots are $x = -3$ and $x = 1$. Evaluating $f$:

$$\begin{aligned} f(-3) &= -9 + 9 + 3 - 1 \\ &= 2 \end{aligned}$$

and

$$\begin{aligned} f(1) &= \frac{1}{3} + 1 - 1 - 1 \\ &= -\frac{2}{3}. \end{aligned}$$

The points are $(-3, 2)$ and $(1, -2/3)$.

**3. Slope 0:** Solve $x^2 + 2x - 1 = 0$. By the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2} \\ &= \frac{-2 \pm \sqrt{8}}{2} \\ &= -1 \pm \sqrt{2} \end{aligned}$$

**Theorem 0.4** (Derivative of natural logarithm)**.** *For every $x > 0$, the derivative of the natural logarithm is:*

$$\begin{aligned}\log'\{x\} &= \frac{1}{x} \\ &= x^{-1}\end{aligned}$$

**Exercise 0.3** (Derivative after logarithmic simplification)**.** Adapted from Miller [10], Question 1.1.4.
Find the derivative of

$$f(x) = \log(4x) - \log(2x)$$

for $x > 0$.

Solution

*Solution* 0.3. Simplify before differentiating using the quotient property of logarithms:

$$\begin{aligned}f(x) &= \log(4x) - \log(2x) \\ &= \log\left(\frac{4x}{2x}\right) \\ &= \log 2\end{aligned}$$

Because $\log 2$ is constant with respect to $x$, the constant rule (Theorem 0.1) gives:

$$f'(x) = 0$$

*Remark:* While one could apply the chain rule separately to each term, giving

$$\begin{aligned}\frac{4}{4x} - \frac{2}{2x} &= \frac{1}{x} - \frac{1}{x} \\ &= 0,\end{aligned}$$

simplifying algebraically first is faster and prevents arithmetic errors.

**Theorem 0.5** (Derivative of exponential)**.** *For every real $x$, the derivative of the exponential function is:*

$$\exp'\{x\} = \exp\{x\}$$

**Theorem 0.6** (Derivative of sine)**.** *For every real $x$, the derivative of the sine function is:*

$$\frac{\partial}{\partial x}\sin x = \cos x$$

**Theorem 0.7** (Derivative of cosine)**.** *For every real $x$, the derivative of the cosine function is:*

$$\frac{\partial}{\partial x}\cos x = -\sin x$$

**Theorem 0.8** (Product rule)**.** *If $a$ and $b$ are differentiable functions of $x$ [11], then*

$$(ab)' = ab' + ba'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Product Rule for Derivatives

**Exercise 0.4** (Derivative of a product)**.** Adapted from Miller [10], Question 1.1.2.
Find the derivative of

$$f(x) = (x^4 + 3x^2 + 8)\cos x$$

Solution

*Solution* 0.4. Apply the product rule (Theorem 0.8) with:

$$u(x) = x^4 + 3x^2 + 8, \qquad v(x) = \cos x$$

Their derivatives are:

$$u'(x) = 4x^3 + 6x, \qquad v'(x) = -\sin x$$

By the product rule:

$$\begin{aligned} f'(x) &= u'(x)v(x) + u(x)v'(x) \\ &= (4x^3 + 6x)\cos x - (x^4 + 3x^2 + 8)\sin x \end{aligned}$$

**Exercise 0.5** (Second derivative of a product)**.** Adapted from Miller [10], Question 1.1.15.
Find the second derivative of

$$f(x) = (x^4 + 3x^2 + 8)\cos x$$

Solution

*Solution* 0.5. From Exercise 0.4, the first derivative is:

$$f'(x) = (4x^3 + 6x)\cos x - (x^4 + 3x^2 + 8)\sin x$$

Differentiate each term via the product rule:

$$\frac{d}{dx}\left[(4x^3 + 6x)\cos x\right] = (12x^2 + 6)\cos x - (4x^3 + 6x)\sin x$$
$$\frac{d}{dx}\left[-(x^4 + 3x^2 + 8)\sin x\right] = -(4x^3 + 6x)\sin x - (x^4 + 3x^2 + 8)\cos x$$

Combining and grouping like trigonometric terms:

$$\begin{aligned} f''(x) &= \left[(12x^2 + 6) - (x^4 + 3x^2 + 8)\right]\cos x - 2(4x^3 + 6x)\sin x \\ &= (-x^4 + 9x^2 - 2)\cos x - (8x^3 + 12x)\sin x \end{aligned}$$

**Theorem 0.9** (Quotient rule)**.** *If $a$ and $b$ are differentiable functions of $x$ [12] and $b \neq 0$, then*

$$(a/b)' = a'/b - (a/b^2)b'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Quotient Rule for Derivatives

**Exercise 0.6** (Derivative of a rational function)**.** Adapted from Miller [10], Question 1.1.12.
Find the derivative of

$$f(x) = \frac{x^2 - 1}{x - 1}$$

for $x \neq 1$.

Solution

*Solution* 0.6. Factor the numerator before differentiating:

$$\begin{aligned} f(x) &= \frac{(x-1)(x+1)}{x-1} \\ &= x+1 \qquad (x \neq 1) \end{aligned}$$

Differentiating directly gives $f'(x) = 1$.
*Alternative (quotient rule):* Using Theorem 0.9 with $u(x) = x^2 - 1$ and $v(x) = x - 1$:

$$\begin{aligned} f'(x) &= \frac{2x(x-1) - (x^2-1)(1)}{(x-1)^2} \\ &= \frac{x^2 - 2x + 1}{(x-1)^2} \\ &= \frac{(x-1)^2}{(x-1)^2} \\ &= 1 \end{aligned}$$

Both methods yield 1, but factoring first eliminates tedious algebraic simplification.

**Theorem 0.10** (Chain rule)**.** *If $a$ is a differentiable function of $b$ [13], and $b$ is a differentiable function of $c$, then $a$ is a differentiable function of $c$, and*

$$\begin{aligned} \frac{da}{dc} &= \frac{da}{db}\frac{db}{dc} \\ &= \frac{db}{dc}\frac{da}{db} \end{aligned}$$

*or in Lagrange's notation (Definition 0.7), if $g$ is differentiable at $x$ and $f$ is differentiable at $g(x)$, then for the composition $f \circ g$, with inner function $g$ and outer function $f$:*

$$(f(g(x)))' = g'(x)f'(g(x))$$

**Exercise 0.7** (Derivative of the standard Gaussian kernel)**.** Adapted from Miller [10], Question 1.1.5.
Find the derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution

*Solution* 0.7. Apply the chain rule (Theorem 0.10) with outer function $\exp\{u\}$ and inner function $u(x) = -x^2/2$. Since $\frac{d}{du}\exp\{u\} = \exp\{u\}$ (Theorem 0.5) and $u'(x) = -x$:

$$\begin{aligned} f'(x) &= u'(x)\exp\{u(x)\} \\ &= -x\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

**Theorem 0.11** (Derivative of the standard Gaussian kernel)**.** *For* $f(x) = \exp\left\{-\frac{x^2}{2}\right\}$ *and every* $x \in \mathbb{R}$,

$$f'(x) = -x\exp\left\{-\frac{x^2}{2}\right\}$$

Proof

*Proof.* This is the solution to Exercise 0.7, worked in Solution 0.7. □

**Exercise 0.8** (Second derivative of the Gaussian kernel)**.** Adapted from Miller [10], Question 1.1.6.
Find the second derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution

*Solution* 0.8. From Theorem 0.11, the first derivative is $f'(x) = -x\exp\{-x^2/2\}$. Apply the product rule (Theorem 0.8) to $u(x) = -x$ and $v(x) = \exp\{-x^2/2\}$:

$$\begin{aligned} u'(x) &= -1 \\ v'(x) &= -x\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

Therefore:

$$
\begin{aligned}
f''(x) &= u'(x)v(x) + u(x)v'(x) \\
&= (-1)\exp\left\{-\frac{x^2}{2}\right\} + (-x)\left[-x\exp\left\{-\frac{x^2}{2}\right\}\right] \\
&= (x^2-1)\exp\left\{-\frac{x^2}{2}\right\}
\end{aligned}
$$

**Theorem 0.12** (Second derivative of the standard Gaussian kernel)**.** *For* $f(x) = \exp\left\{-\frac{x^2}{2}\right\}$ *and every* $x \in \mathbb{R}$*,*

$$
f''(x) = (x^2-1)\exp\left\{-\frac{x^2}{2}\right\}
$$

Proof

*Proof.* This is the solution to Exercise 0.8, worked in Solution 0.8. □

**Exercise 0.9** (Combining product and chain rules)**.** Adapted from Miller [10], Question 1.1.7.
Find the derivative of

$$
f(x) = \exp\{x^8\}\cos(3x^4)
$$

Solution

*Solution* 0.9. Apply the product rule (Theorem 0.8) to $u(x) = \exp\{x^8\}$ and $v(x) = \cos(3x^4)$:

$$
f'(x) = u'(x)v(x) + u(x)v'(x)
$$

Compute $u'(x)$ and $v'(x)$ via the chain rule (Theorem 0.10):

$$
\begin{aligned}
u'(x) &= 8x^7\exp\{x^8\} \\
v'(x) &= -12x^3\sin(3x^4)
\end{aligned}
$$

Substitute both derivatives into the product rule:

$$
\begin{aligned}
f'(x) &= 8x^7\exp\{x^8\}\cos(3x^4) - 12x^3\exp\{x^8\}\sin(3x^4) \\
&= 4x^3\exp\{x^8\}[2x^4\cos(3x^4) - 3\sin(3x^4)]
\end{aligned}
$$

**Exercise 0.10** (Generalized power rule)**.** Adapted from Miller [10], Question 1.1.13. Find the derivative of

$$\begin{aligned} f(x) &= \sqrt[3]{(5x-2)^2} \\ &= (5x-2)^{2/3} \end{aligned}$$

for $x \neq 2/5$.

Solution

*Solution* 0.10. Apply the generalized power rule $\frac{d}{dx}[g(x)]^r = r[g(x)]^{r-1}g'(x)$ with $g(x) = 5x - 2$ and $r = 2/3$:

$$\begin{aligned} f'(x) &= \frac{2}{3}(5x-2)^{2/3-1} \cdot 5 \\ &= \frac{10}{3}(5x-2)^{-1/3} \\ &= \frac{10}{3(5x-2)^{1/3}} \end{aligned}$$

*Remark:* Omitting the inner derivative $g'(x) = 5$ is a frequent mistake when applying the generalized power rule.

**Corollary 0.1** (Chain rule for logarithms)**.** *If $f$ is differentiable at $x$ and $f(x) > 0$, then*

$$\frac{d}{dx}\log\{f(x)\} = \frac{f'(x)}{f(x)}$$

Proof

*Proof.* Apply Theorem 0.10 and Theorem 0.4:

$$\begin{aligned} \frac{d}{dx}\log\{f(x)\} &= f'(x) \cdot \log'\{f(x)\} && \text{(chain rule, with } g = f \text{ and outer function } \log\text{)} \\ &= f'(x) \cdot \frac{1}{f(x)} && \text{(derivative of } \log\text{, valid because } f(x) > 0\text{)} \\ &= \frac{f'(x)}{f(x)} && \text{(multiply)} \end{aligned}$$

□

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule and on exercises that combine it with the other rules:

- The Chain Rule for Derivatives
- The Power Rule on a Function Chain
- Advanced Exercises on Derivative Rules

**Exercise 0.11** (Chain rule with logarithm)**.** Adapted from Miller [10], Question 1.1.3. Find the derivative of

$$f(x) = \log(1 - x^2)$$

for $x \in (-1, 1)$.

Solution

*Solution* 0.11. Apply the chain rule for logarithms (Corollary 0.1): if $g(x) = 1 - x^2$, then $g'(x) = -2x$, and

$$\begin{aligned} f'(x) &= \frac{g'(x)}{g(x)} \\ &= -\frac{2x}{1 - x^2} \end{aligned}$$

*Remark:* A common pitfall in chain rule problems is evaluating the outer derivative at $x$ rather than at the inner value $g(x)$. The denominator is $g(x) = 1 - x^2$, not $x$.

**Exercise 0.12** (Second derivative with an added constant)**.** Adapted from Miller [10], Question 1.1.9.
Find the second derivative of

$$f(x) = \log x + \sqrt{162}$$

for $x > 0$.

Solution

*Solution* 0.12. Because $\sqrt{162}$ is constant with respect to $x$, its derivative is zero (Theorem 0.1):

$$\begin{aligned} f'(x) &= \frac{1}{x} \\ &= x^{-1} \end{aligned}$$

Differentiating again via the power rule (Theorem 0.3):

$$\begin{aligned} f''(x) &= -x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

## Studying a new function

**Definition 0.1** (Logarithmic derivative)**.** The **logarithmic derivative** $\frac{f'(x)}{f(x)}$ of a function $f$ of a real variable equals $\frac{d}{dx}\log\{f(x)\}$ by Corollary 0.1 and measures the rate of change of $f$ relative to its size; it needs $f(x) > 0$, and for $f(x) < 0$ the same expression is the derivative of $\log\{|f(x)|\}$.

**Definition 0.2** (Profile of a function)**.** The **profile** of a function $f$ of a real variable is the list of related functions and values to work out when meeting $f$ for the first time:

- the logarithm $\log\{f(x)\}$, where $f(x) > 0$;
- the derivatives $f'(x)$, $f''(x)$, and so on, as far as they are informative;
- the logarithmic derivative $\frac{f'(x)}{f(x)}$;
- the inverse function $f^{-1}$, which exists on the image of $f$ when $f$ is one-to-one, and otherwise only after restricting the domain of $f$;
- the complement $1 - f(x)$, which is informative when $f$ takes values in $[0, 1]$, as a probability does;
- the reciprocal $\frac{1}{f(x)}$, where $f(x) \neq 0$ (not to be confused with $f^{-1}$);
- an antiderivative of $f$, a function $F$ with $F' = f$ (integration);
- the values $f(0)$ and $f(1)$, and the limits of $f(x)$ as $x \to \infty$ and as $x \to -\infty$, at each of these that lies in the domain of $f$ or exists;
- the graph of $f$.

**Example 0.1** (The profile of the logistic function)**.** The **logistic function** (also called expit or the sigmoid) is

$$f(x) = \frac{1}{1 + e^{-x}}.$$

The profile in Definition 0.2 of $f$ is as follows. Write $E = e^{-x}$, so that $f = \frac{1}{1+E}$ and $1 - f = \frac{E}{1+E}$.

- **Logarithm:** $\log\{f(x)\} = -\log\{1 + e^{-x}\}$, defined for every real $x$ because $f(x) > 0$.
- **Derivative:** by the chain rule,

$$\begin{aligned} f'(x) &= \frac{e^{-x}}{(1 + e^{-x})^2} && \text{(chain rule applied to } (1 + e^{-x})^{-1}) \\ &= \frac{1}{1 + E} \cdot \frac{E}{1 + E} && \text{(split the square)} \\ &= f(x)\,(1 - f(x)). && \text{(identify } f \text{ and } 1 - f) \end{aligned}$$

  Differentiating again with the product rule gives $f''(x) = f'(1 - f) - ff' = f(x)(1 - f(x))(1 - 2f(x))$.
- **Logarithmic derivative:** $\frac{f'(x)}{f(x)} = 1 - f(x)$, so the relative rate of change of $f$ is the complement of $f$ itself.
- **Inverse:** for $y \in (0, 1)$, solving $y = \frac{1}{1+e^{-x}}$ for $x$ gives

$$\begin{aligned} \frac{1}{y} &= 1 + e^{-x} \\ e^{-x} &= \frac{1 - y}{y} \\ x &= \log\left\{\frac{y}{1 - y}\right\}. \end{aligned}$$

  So $f^{-1}(y) = \text{logit}\{y\}$, the log-odds. The inverse exists because $f' = f(1 - f) > 0$, so $f$ is strictly increasing and therefore one-to-one.
- **Complement:** $1 - f(x) = \frac{e^{-x}}{1+e^{-x}} = \frac{1}{e^{x}+1} = f(-x)$.
- **Reciprocal:** $\frac{1}{f(x)} = 1 + e^{-x}$.
- **Antiderivative:** $F(x) = \log\{1 + e^{x}\}$ satisfies $F'(x) = \frac{e^{x}}{1+e^{x}} = f(x)$, so $\int f(x)\,dx = \log\{1 + e^{x}\} + C$.
- **Key values:** $f(0) = \frac{1}{1+1} = \frac{1}{2}$ and $f(1) = \frac{1}{1+e^{-1}}$. As $x \to \infty$, $e^{-x} \to 0$, so $f(x) \to 1$; as $x \to -\infty$, $e^{-x} \to \infty$, so $f(x) \to 0$.

The code checks these results numerically, using the base R functions `plogis()` for $f$ and `qlogis()` for $f^{-1}$, and `D()` to differentiate symbolically.

```
f <- plogis
x <- c(-2, -0.5, 0, 1, 3)

f_expr <- quote(1 / (1 + exp(-x)))
f_prime <- eval(D(f_expr, "x"), list(x = x))
f_second <- eval(D(D(f_expr, "x"), "x"), list(x = x))

checks <- c(
  derivative = max(abs(f_prime - f(x) * (1 - f(x)))),
  second_derivative = max(abs(f_second - f(x) * (1 - f(x)) * (1 - 2 * f(x)))),
  log_derivative = max(abs(f_prime / f(x) - (1 - f(x)))),
  inverse = max(abs(qlogis(f(x)) - x)),
  complement = max(abs(1 - f(x) - f(-x))),
  reciprocal = max(abs(1 / f(x) - (1 + exp(-x)))),
  antiderivative = abs(
    integrate(f, 0, 1)$value - (log(1 + exp(1)) - log(2))
  )
)
signif(checks, 3)
#>        derivative second_derivative    log_derivative           inverse
#>          1.46e-16          1.18e-16          1.53e-16          3.11e-15
#>        complement        reciprocal    antiderivative
#>          1.46e-16          0.00e+00          0.00e+00
stopifnot(all(checks < 1e-14))

key_values <- c(f(0), f(1), f(10), f(-10))
```

Each entry of `checks` is the largest discrepancy between a formula above and a numerical computation; every one is below $10^{-14}$ (the chunk stops with an error otherwise), which is rounding error. The key values are $f(0) = 0.5$ and $f(1) \approx 0.7311$. At $x = 10$ and $x = -10$ the function is already close to its limits, $f(10) \approx 0.99995$ and $f(-10) \approx 0.0000454$. Figure 1 graphs $f$, $f'$ and $\log\{f\}$.

```r
grid <- seq(-6, 6, length.out = 241)
panels <- data.frame(
  x = rep(grid, 3),
  panel = factor(
    rep(c("f(x)", "f'(x)", "log f(x)"), each = length(grid)),
    levels = c("f(x)", "f'(x)", "log f(x)")
  ),
  value = c(f(grid), f(grid) * (1 - f(grid)), log(f(grid)))
)
ggplot2::ggplot(panels) +
  ggplot2::aes(x = x, y = value) +
  ggplot2::geom_line() +
  ggplot2::facet_wrap(~panel, scales = "free_y") +
  ggplot2::labs(x = "Input, x", y = NULL)
```

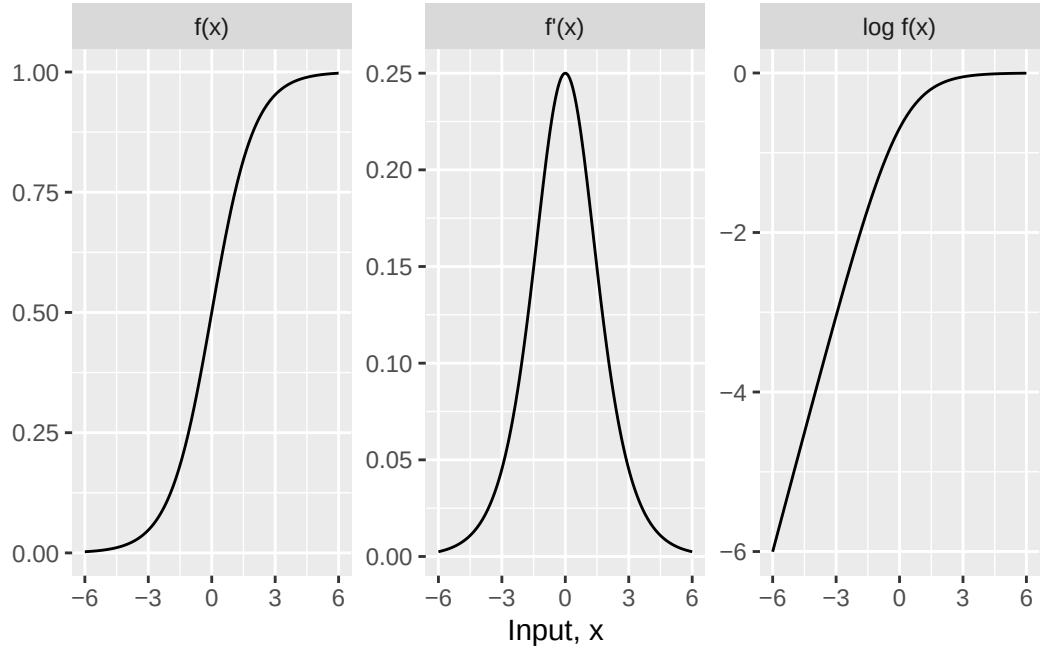



Figure 1: The logistic function, its derivative and its logarithm

## Tangent lines

**Definition 0.1** (Tangent line)**.** Let $f$ be differentiable at $c$ (Definition 0.6). The **tangent line** to the graph of $f$ at $c$ is the graph of the affine function

$$x \mapsto f(c) + f'(c)\,(x - c),$$

the line through the point $(c, f(c))$ with slope $f'(c)$. The slope $f'(c)$ is also called the **tangent slope** of $f$ at $c$.

**Example 0.1** (The tangent line to $x^2$ at 1)**.** For $f(x) = x^2$ at $c = 1$, $f(1) = 1$ and $f'(1) = 2$ (Example 0.5), so the tangent line is the graph of

$$\begin{aligned} x &\mapsto 1 + 2\,(x - 1) && \text{(substitute } f(1) = 1 \text{ and } f'(1) = 2\text{)} \\ &= 2x - 1 && \text{(distribute, and } 1 - 2 = -1\text{)} \end{aligned}$$

At $x = 1.1$ the tangent line has height $2(1.1) - 1 = 1.2$, close to the curve's height $f(1.1) = 1.21$; the gap, $1.21 - 1.2 = 0.01$, is $(1.1 - 1)^2$.

## Linear approximation

For a differentiable function $f$ and a small step $\varepsilon$,

$$f(w + \varepsilon) \approx f(w) + \varepsilon \frac{d}{dw} f(w) \tag{1}$$

**Definition 0.1** (Linear approximation (first-order Taylor approximation, first-order Taylor expansion))**.** For a function $f$ that is differentiable at $w$, the **linear approximation** of $f$ at $w$ [14] (also called the **first-order Taylor approximation** or the **first-order Taylor expansion**) is the function of the step $\varepsilon$

$$\hat{f}_w(\varepsilon) = f(w) + \varepsilon \frac{d}{dw} f(w)$$

so Equation 1 says $f(w + \varepsilon) \approx \hat{f}_w(\varepsilon)$ for small $\varepsilon$. Figure 2 illustrates this approximation interactively for a quadratic function.

Remark (The "linear" approximation is affine)

*Remark* 0.1 (The "linear" approximation is affine). The linear approximation $\hat{f}_w$ (Definition 0.1) is an affine function of the step $\varepsilon$: its slope is $\frac{d}{dw}f(w)$ and its intercept is $f(w)$. It is linear in $\varepsilon$ only when $f(w) = 0$. The name "linear approximation" uses "linear" in the looser sense of elementary algebra (remark). Boyd and Vandenberghe [15], section 2.2, treats the first-order Taylor approximation as an affine function for functions of several variables as well.

*(This plot is interactive; see the HTML or slide version of this page to view it.)*

Figure 2: The linear approximation Equation 1 for $f(w) = w^2 - 4w + 7$, at any $w$ and step $\varepsilon$.

**Exercise 0.1** (Tangent line approximation)**.** Adapted from Miller [10], Question 1.1.8.
Let $f(x) = 4x + \sqrt{2}\cos x$.
Find $f'(x)$.

Solution

*Solution* 0.1. Differentiate each term:

$$f'(x) = 4 - \sqrt{2}\sin x$$

**Exercise 0.2** (Tangent line at a point)**.** In the setting of Exercise 0.1, find the equation of the tangent line to the curve $y = f(x)$ at $x = \pi/4$.

Solution

*Solution* 0.2. Evaluate $f$ and $f'$ at $x_0 = \pi/4$:

$$
\begin{aligned}
f(\pi/4) &= 4\left(\frac{\pi}{4}\right) + \sqrt{2}\cos\left(\frac{\pi}{4}\right) \\
&= \pi + \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\
&= \pi + 1 \\
f'(\pi/4) &= 4 - \sqrt{2}\sin\left(\frac{\pi}{4}\right) \\
&= 4 - \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\
&= 4 - 1 \\
&= 3
\end{aligned}
$$

Using the point-slope formula, the tangent line at $(\pi/4, \pi + 1)$ is:

$$
y - (\pi + 1) = 3\left(x - \frac{\pi}{4}\right) \implies y = \pi + 1 + 3\left(x - \frac{\pi}{4}\right)
$$

**Exercise 0.3** (Approximate with the tangent line)**.** Use the tangent line from Exercise 0.2 to approximate $f(\pi/4 + 0.01)$.

Solution

*Solution* 0.3. At $x = \pi/4 + 0.01$, the linear approximation (Equation 1) gives:

$$
f(\pi/4 + 0.01) \approx (\pi + 1) + 3(0.01) = \pi + 1.03 \approx 4.171593
$$

**Exercise 0.4** (Check the approximation)**.** Compare the approximation in Exercise 0.3 with the exact value of $f(\pi/4 + 0.01)$.

Solution

*Solution* 0.4. The exact value is:

$$
f(\pi/4 + 0.01) = 4\left(\frac{\pi}{4} + 0.01\right) + \sqrt{2}\cos\left(\frac{\pi}{4} + 0.01\right) \approx 4.171543
$$

The approximation error is $|4.171593 - 4.171543| \approx 0.00005$, on the order of $(0.01)^2 = 10^{-4}$.

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